Doiraning yuzi

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. Halqaning tashqi diametri 22 cm, ichki radiusi 6 cm. Halqaning yuzini toping (π = 3,14).

A) 1406,72 cm² **B)** 266,9 cm²
C) 78,5 cm² **D)** 492,98 cm²

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2. Dumaloq sahnaning radiusi 9 m. Sahna atrofiga eni 2 m bo'lgan gilam to'shaldi. Gilam qancha joyni egallaydi? (π = 3,14)

A) 379,94 m² **B)** 56,52 m² **C)** 12,56 m² **D)** 125,6 m²

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3. Tomoni 18 dm bo'lgan kvadrat tunuka listidan eng katta dumaloq qopqoq qirqildi. Qancha tunuka chiqindiga ketdi? (π = 3,14)

A) 254,34 dm² **B)** 306,96 dm²
C) 17,04 dm² **D)** 69,66 dm²

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4. Radiusi 10 cm bo'lgan dumaloq tort ustiga kvadrat shokolad plitka qo'yildi: plitkaning to'rtala burchagi tort chetiga tegib turadi. Tortning plitka bilan yopilmagan qismi yuzini toping (π = 3,14).

A) 86 cm² **B)** 114 cm² **C)** 214 cm² **D)** 14 cm²

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5. Dumaloq stol uchun diametri 2 m bo'lgan shisha qirqildi. 1 m² shisha 150 000 so'm turadi. Shisha necha so'm turadi? (π = 3,14)

A) 942 000 so'm **B)** 1 884 000 so'm
C) 300 000 so'm **D)** 471 000 so'm

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6. Diametri 40 cm bo'lgan bitta pitsa va diametri 20 cm bo'lgan to'rtta pitsa bor. Qaysi birida yeyiladigan joy ko'p? (π = 3,14)

A) to'rtta kichik pitsada **B)** ikkalasida teng
C) bitta katta pitsada **D)** katta pitsada 2 baravar ko'p

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7. Sirk arenasi — radiusi 10,5 m bo'lgan doira. Unga har 1 m² ga 20 kg qum sepiladi. Necha kilogramm qum kerak? (π = $\frac{22}{7}$)

A) 6930 kg **B)** 1320 kg **C)** 3465 kg **D)** 27 720 kg

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8. Radiusi 2,7 cm bo'lgan doiraning yuzini toping (π = 3,14). Javobni yuzdan birlar xonasigacha yaxlitlang.

A) 16,96 cm² **B)** 22,89 cm² **C)** 22,88 cm² **D)** 22,9 cm²

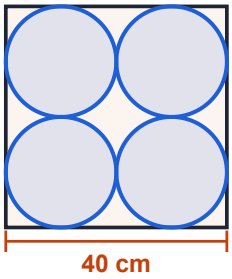
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9. Diametri 7 m bo'lgan doiraning yuzini toping (π = 3,14). Javobni o'ndan birlar xonasigacha yaxlitlang.

A) 22,0 m² **B)** 38,4 m² **C)** 153,9 m² **D)** 38,5 m²

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10. **Tomoni 40 cm bo'lgan kvadrat patnisga 4 ta bir xil dumaloq non rasmdagidek qo'yildi: nonlar bir-biriga va patnis chetlariga tegib turadi. Patnisning nonlar bilan yopilmagan qismi yuzini toping (π = 3,14).**



A) 344 cm²
B) 1256 cm²
C) 86 cm²
D) 1286 cm²

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11. **Radiusi 6 cm bo'lgan doiraning yuzi radiusi 3 cm bo'lgan doiraning yuzidan necha marta katta?**

A) 3 marta
B) 2 marta
C) 6 marta
D) 4 marta

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12. **Halqaning tashqi radiusi 8 cm, ichki radiusi 3 cm. Halqaning yuzini π orqali ifodalang.**

A) 73π cm²
B) 25π cm²
C) 55π cm²
D) 10π cm²

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13. **Doira shaklidagi gulzorning diametri 6 m. Uning o'rtasida diametri 2 m bo'lgan dumaloq favvora bor. Gul ekiladigan joyning yuzini toping (π = 3,14).**

A) 100,48 m²
B) 25,12 m²
C) 12,56 m²
D) 31,4 m²

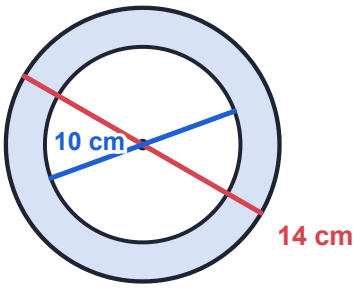
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14. **Diametri 0,8 m bo'lgan doiraning yuzini π orqali ifodalang.**

A) 0,64π m²
B) 0,16π m²
C) 1,6π m²
D) 0,8π m²

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15. **Rasmda halqa berilgan: tashqi diametri 14 cm, ichki diametri 10 cm. Halqaning yuzini toping (π = 3,14).**



A) 12,56 cm²
B) 50,24 cm²
C) 75,36 cm²
D) 301,44 cm²

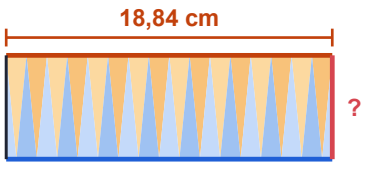
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16. **Dumaloq qopqoqning radiusi 0,5 m. Uning ikkala tomoni bo'yaladi. Har 1 m² ga 300 g bo'yoq ketadi. Necha gramm bo'yoq kerak? (π = 3,14)**

A) 235,5 g
B) 471 g
C) 1884 g
D) 942 g

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17. **Doira bo'laklaridan rasmdagidek deyarli to'g'ri to'rtburchak tuzildi. Uning eni 18,84 cm. Doiraning yuzini toping (π = 3,14).**



A) 18,84 cm²
B) 354,95 cm²
C) 37,68 cm²
D) 113,04 cm²

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18. **A doiraning radiusi 4 cm, B doiraning aylana uzunligi 31,4 cm. Qaysi doiraning yuzi katta va qanchaga? (π = 3,14)**

A) B doiraniki 6,28 cm² ga katta
B) A doiraniki 28,26 cm² ga katta
C) ikkalasi teng
D) B doiraniki 28,26 cm² ga katta

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19. Doiraning radiusi 2,5 cm. Lola uning yuzini 3,14 · 2,5 · 2 = 15,7 cm², Kamola esa 3,14 · 5 · 5 = 78,5 cm² deb topdi. Kim to'g'ri topgan?

A) Lola

B) Kamola

C) ikkalasi ham xato, javob 19,625 cm²

D) ikkalasi ham to'g'ri

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20. Ikkita doiraning radiuslari 5 cm va 7 cm. Ularning yuzlari yig'indisini toping (π = $\frac{22}{7}$).

A) $452\frac{4}{7}$ cm²

B) $75\frac{3}{7}$ cm²

C) $232\frac{4}{7}$ cm²

D) 154 cm²

Javoblar

Doiraning yuzi · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	266,9 cm²
2	**D**	125,6 m²
3	**D**	69,66 dm²
4	**B**	114 cm²
5	**D**	471 000 so'm
6	**B**	ikkalasida teng
7	**A**	6930 kg
8	**B**	22,89 cm²
9	**D**	38,5 m²
10	**A**	344 cm²

№	Javob	Qiymati
11	**D**	4 marta
12	**C**	55π cm²
13	**B**	25,12 m²
14	**B**	0,16π m²
15	**C**	75,36 cm²
16	**B**	471 g
17	**D**	113,04 cm²
18	**D**	B doiraniki 28,26 cm² ga katta
19	**C**	ikkalasi ham xato, javob 19,625 cm²
20	**C**	$232\frac{4}{7}$ cm²

Qisqa yechimlar

1. Tashqi radius: 22 : 2 = 11 cm. Katta doira: 3,14 · 121 = 379,94 cm².
Kichik doira: 3,14 · 36 = 113,04 cm².
Halqa: 379,94 − 113,04 = **266,9 cm²**.

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2. Katta doiraning radiusi: 9 + 2 = 11 m. Yuzi: 3,14 · 121 = 379,94 m².
Sahna: 3,14 · 81 = 254,34 m².
Gilam: 379,94 − 254,34 = **125,6 m²**.

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3. Doira diametri — 18 dm, radiusi 9 dm. Doira: 3,14 · 81 = 254,34 dm².
Kvadrat: 18 · 18 = 324 dm².
Chiqindi: 324 − 254,34 = **69,66 dm²**.

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4. Tort: 3,14 · 10 · 10 = 314 cm².
Plitka (kvadrat) 4 ta uchburchakdan iborat, har biri 10 · 10 : 2 = 50 cm²: 4 · 50 = 200 cm².
Yopilmagan qism: 314 − 200 = **114 cm²**.

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5. Radius: 2 : 2 = 1 m. Yuzi: 3,14 · 1 · 1 = 3,14 m².
Narxi: 3,14 · 150 000 = **471 000 so'm**.

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6. Katta pitsa: r = 20 cm, 3,14 · 400 = 1256 cm².
Kichik pitsa: r = 10 cm, 3,14 · 100 = 314 cm²; to'rttasi: 4 · 314 = 1256 cm².
Javob: **ikkalasida teng**.

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7. Yuzi: $\frac{22}{7}$ · 10,5 · 10,5. Avval 10,5 : 7 = 1,5; 22 · 1,5 · 10,5 = 346,5 m².
Qum: 346,5 · 20 = **6930 kg**.

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8. 2,7 · 2,7 = 7,29. Yuzi: 3,14 · 7,29 = 22,8906 cm².
Uchinchi raqam 0 — yaxlitlaymiz: **22,89 cm²**.

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9. Radius: 7 : 2 = 3,5 m. Yuzi: 3,14 · 3,5 · 3,5 = 38,465 m².
Ikkinchi raqam 6 — 5 dan katta: **38,5 m²**.

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10. Har nonning diametri 40 : 2 = 20 cm, radiusi 10 cm. Bitta non: 314 cm², to'rttasi: 1256 cm².
Patnis: 40 · 40 = 1600 cm².
Yopilmagan qism: 1600 − 1256 = **344 cm²**.

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11. Katta doira: $\pi \cdot 6 \cdot 6 = 36\pi$. Kichik doira: $\pi \cdot 3 \cdot 3 = 9\pi$.
$36\pi : 9\pi =$ **4 marta**. (Radius 2 marta, yuz 4 marta katta.)

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12. Katta doira: $\pi \cdot 8 \cdot 8 = 64\pi$. Kichik doira: $\pi \cdot 3 \cdot 3 = 9\pi$.
Halqa: $64\pi - 9\pi =$ **55π cm²**.
(25π — radiuslar ayirmasi kvadratga oshirilgan, bu noto'g'ri.)

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13. Gulzor: r = 3 m, $3{,}14 \cdot 9 = 28{,}26$ m².
Favvora: r = 1 m, $3{,}14 \cdot 1 = 3{,}14$ m².
Gul joyi: $28{,}26 - 3{,}14 =$ **25,12 m²**.

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14. Radius: $0{,}8 : 2 = 0{,}4$ m.
Yuzi = $\pi \cdot 0{,}4 \cdot 0{,}4 =$ **$0{,}16\pi$ m²**.

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15. Radiuslar: $14 : 2 = 7$ cm va $10 : 2 = 5$ cm.
Katta doira: $3{,}14 \cdot 49 = 153{,}86$ cm². Kichik doira: $3{,}14 \cdot 25 = 78{,}5$ cm².
Halqa: $153{,}86 - 78{,}5 =$ **75,36 cm²**.

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16. Bir tomoni: $3{,}14 \cdot 0{,}5 \cdot 0{,}5 = 0{,}785$ m². Ikkala tomoni: $2 \cdot 0{,}785 = 1{,}57$ m².
Bo'yoq: $1{,}57 \cdot 300 =$ **471 g**.

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17. Eni — aylana uzunligining yarmi: $\pi r = 18{,}84$, demak $r = 18{,}84 : 3{,}14 = 6$ cm. Bo'yi — radius, 6 cm.
Yuzi: $18{,}84 \cdot 6 =$ **113,04 cm²** (yoki $3{,}14 \cdot 6 \cdot 6$).

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18. B: diametri $31{,}4 : 3{,}14 = 10$ cm, radiusi 5 cm; yuzi $3{,}14 \cdot 25 = 78{,}5$ cm².
A: $3{,}14 \cdot 16 = 50{,}24$ cm².
$78{,}5 - 50{,}24 = 28{,}26$: **B doiraniki 28,26 cm² ga katta**.

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19. Lola r^2 o'rniga $2r$ ni oldi, Kamola radius o'rniga diametrni oldi.
To'g'risi: $3{,}14 \cdot 2{,}5 \cdot 2{,}5 = 3{,}14 \cdot 6{,}25 =$ **19,625 cm²**.

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20. Yuzlar: $\frac{22}{7} \cdot 25 = \frac{550}{7}$ cm² va $\frac{22}{7} \cdot 49 = 154$ cm².
Yig'indi: $\frac{550}{7} + 154 = \frac{1628}{7} = 232\frac{4}{7}$ **cm²**.
